Institute of Technology of Cambodia
Faculty of Electrical Engineering
Génie Informatique et Communication

GICI5CC · CLOUD COMPUTING · WEEK 11

Cost Management

AWS payment models, the Free Tier, total cost of ownership and the cost management tools.



Heng Rathpisey · Department GIC · Academic year 2025–2026

Recap

LAST SESSION

Security

The principle of least privilege and the shared responsibility model.

LAST SESSION

AWS IAM

Users, credentials and policies on AWS.

Learning objectives

1. Describe AWS cost and Free Tier structures.
2. Describe the AWS cost management tools that are available.

THIS WEEK

Cost management

The AWS pricing models, the AWS Free Tier, and the pricing tools in the AWS Management Console.

Introduction to AWS cost management



The AWS Cost Management tools give you visibility into your costs and usage, so you can access, organize, understand, control and optimize them.

PART 1

Paying for AWS

Pay only for the individual services you need, for as long as you use them.

AWS payment models



No long-term contracts or complex licensing. Organizations stay agile and responsive, and can always meet scaling demands.

AWS payment models: three ways to pay

01

Pay-as-you-go

Adapt to changing business needs without overcommitting budgets. You plan on need, not on forecasts, which reduces the risk of overprovisioning or missing capacity.

02

Save when you commit

Significant savings on your AWS usage. Sign up for Savings Plans for a 1- or 3-year term, and manage them with recommendations, reporting and budget alerts in AWS Cost Explorer.

03

Pay less by using more

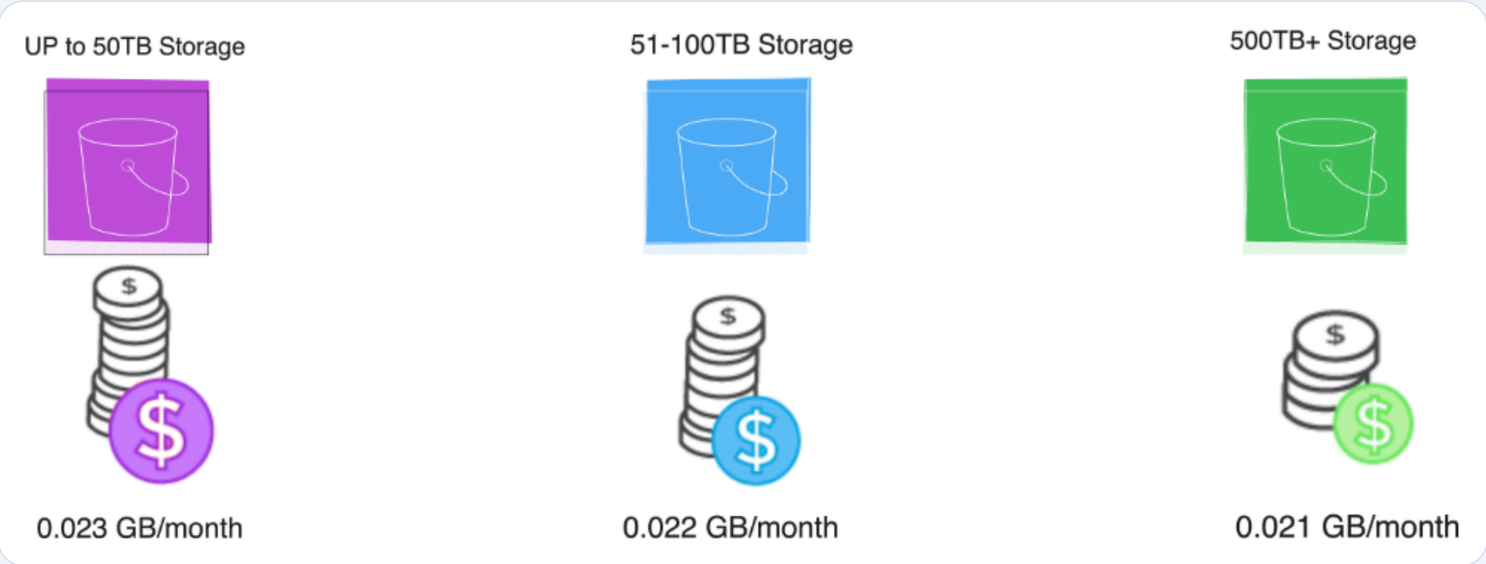
Volume-based discounts as your usage increases. You benefit from economies of scale that keep costs under control as adoption grows.

Save when you commit: an example

EC2 commitment	Monthly cost	Discount
None (pay-as-you-go)	$734.67	0%
1-year commitment	$487.07	33.07%
3-year commitment	$332.15	54.78%

The same EC2 usage costs less than half with a 3-year commitment.

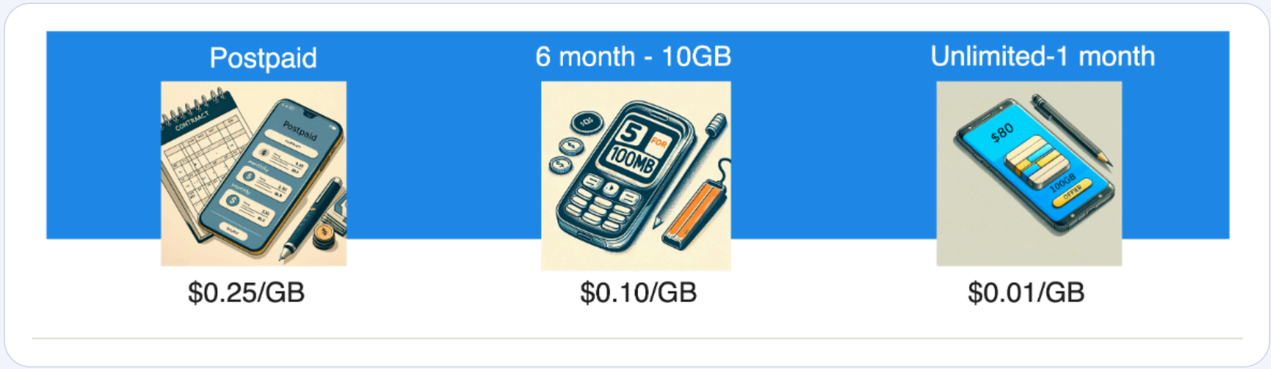
Pay less by using more: storage



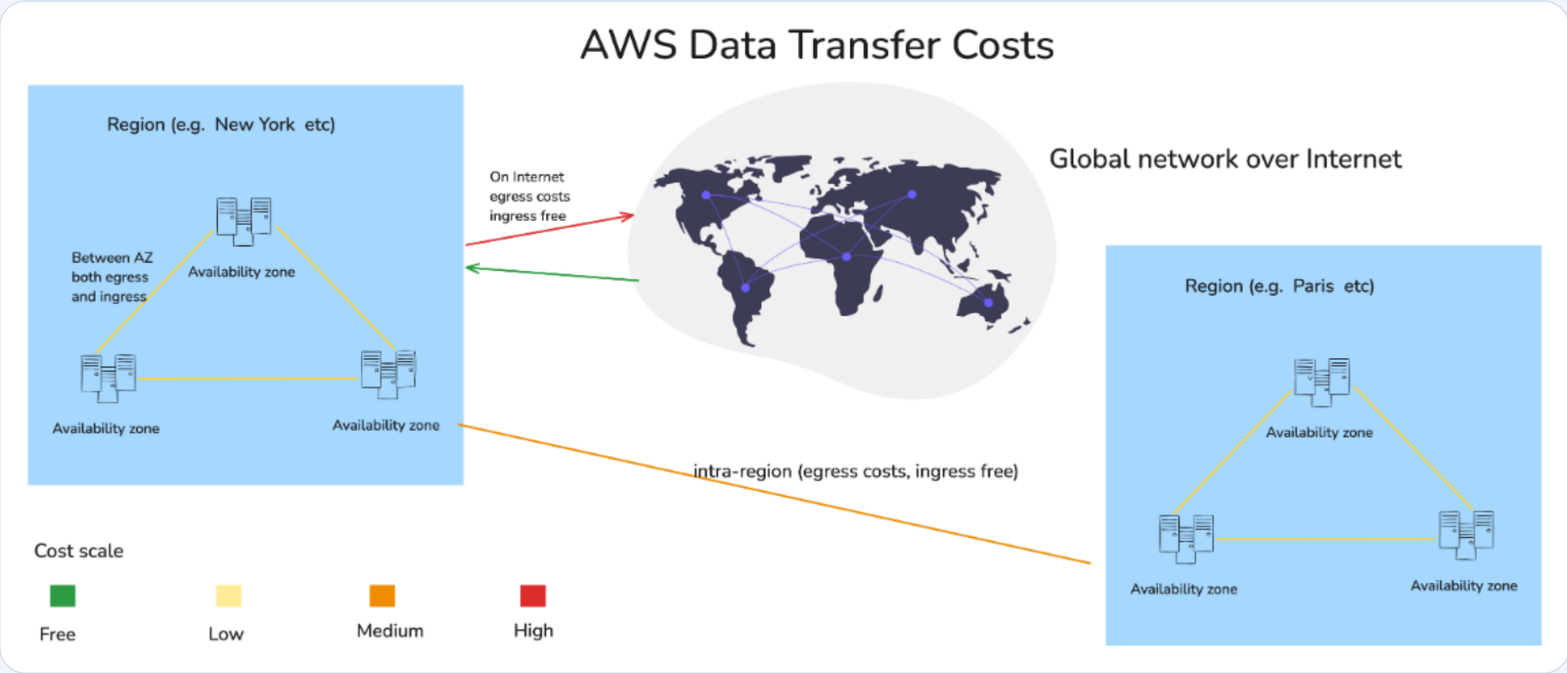
Volume discounts decrease the cost per GB with more use. Economies of scale apply, like wholesale versus retail pricing.

Payment models compared

Feature	Pay-as-you-go	Save when you commit	Pay less by using more
Flexibility	Very high	Low	Medium
Cost savings	No upfront savings	High savings	Volume-based savings
Budget predictability	Low	High	Medium



Bandwidth costs



Transferring data across AWS Regions costs money, and it can be a significant amount.

Bandwidth costs: four rules

Inbound

All inbound traffic from the internet or another Region is free.

Between zones

Transfer between Availability Zones in the same Region costs money in both directions, at the lowest price tier.

Region to Region

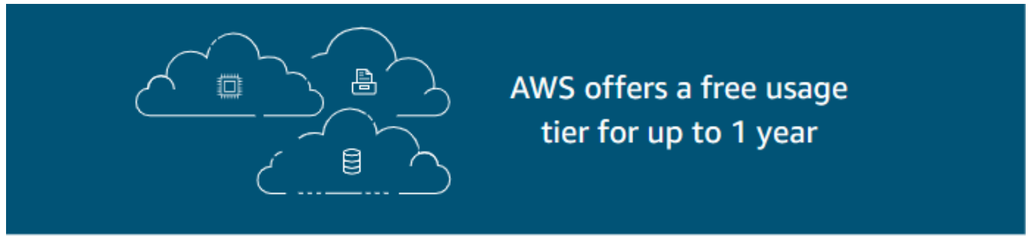
Egress from one Region to another costs more than transfer between zones, but less than egress to the internet.

To the internet

Internet outbound traffic has the highest data costs.

An Availability Zone is a cluster of datacenter buildings. Multiple zones over a metropolitan area form a Region.

AWS Free Tier



Save money as you learn and experiment with AWS Free Tier

Learn and experiment at no cost



Free trials, 12 months free, always free

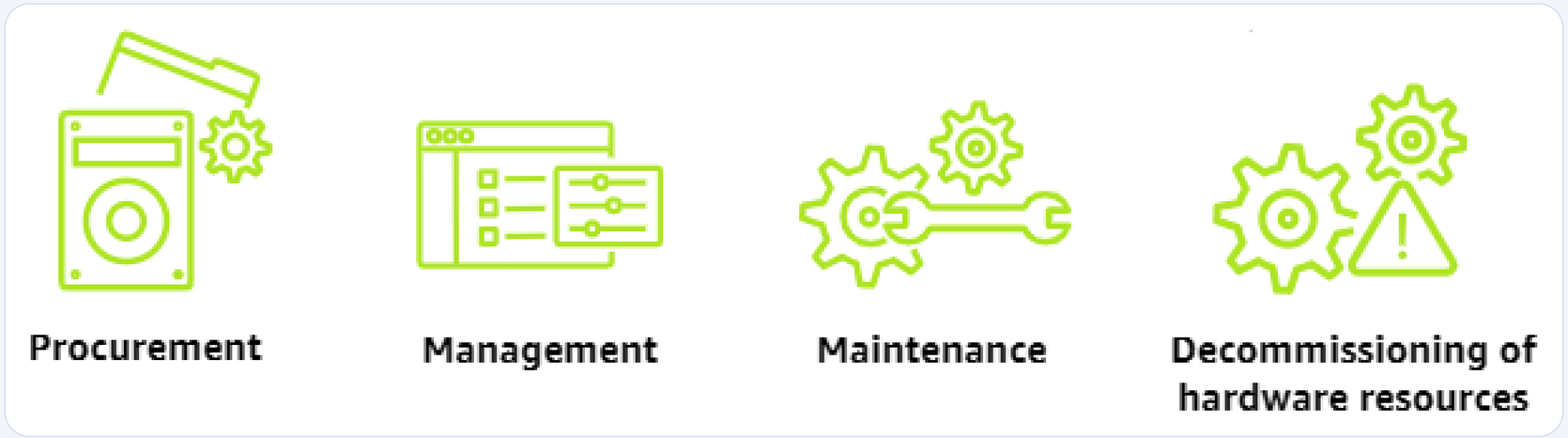
PART 2

Total cost of ownership

A financial metric to estimate and compare the direct and indirect costs of a product or a service.

Week 11 · 14

Total cost of ownership (TCO)



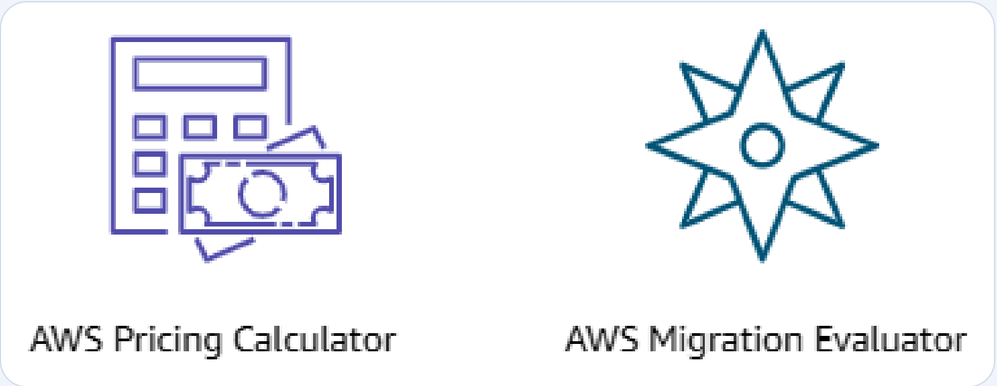
TCO typically includes procurement, management, maintenance and decommissioning of hardware, over a three- or five-year period.

TCO tools

With the variety of hardware and service configurations available today, it can be difficult to understand the total cost.

It is challenging to come up with an accurate TCO model that represents the true cost of running your application.

To help manage TCO, AWS offers pricing tools like the AWS Pricing Calculator and the AWS Migration Evaluator.

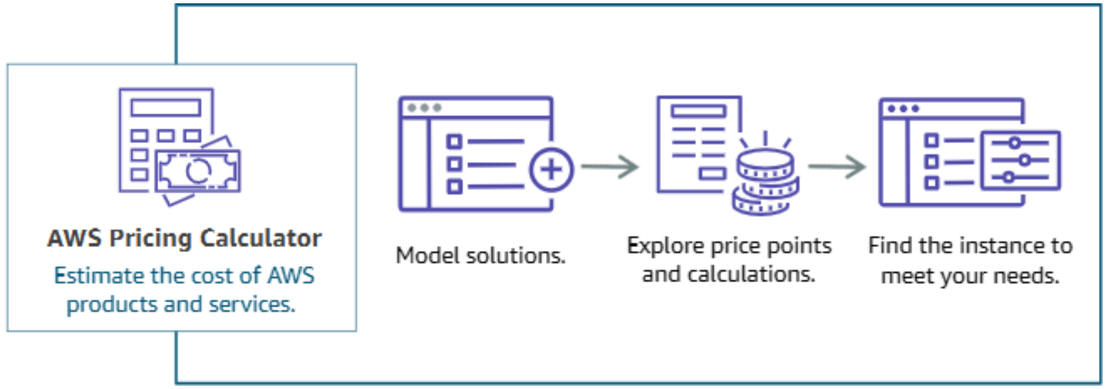


AWS Pricing Calculator

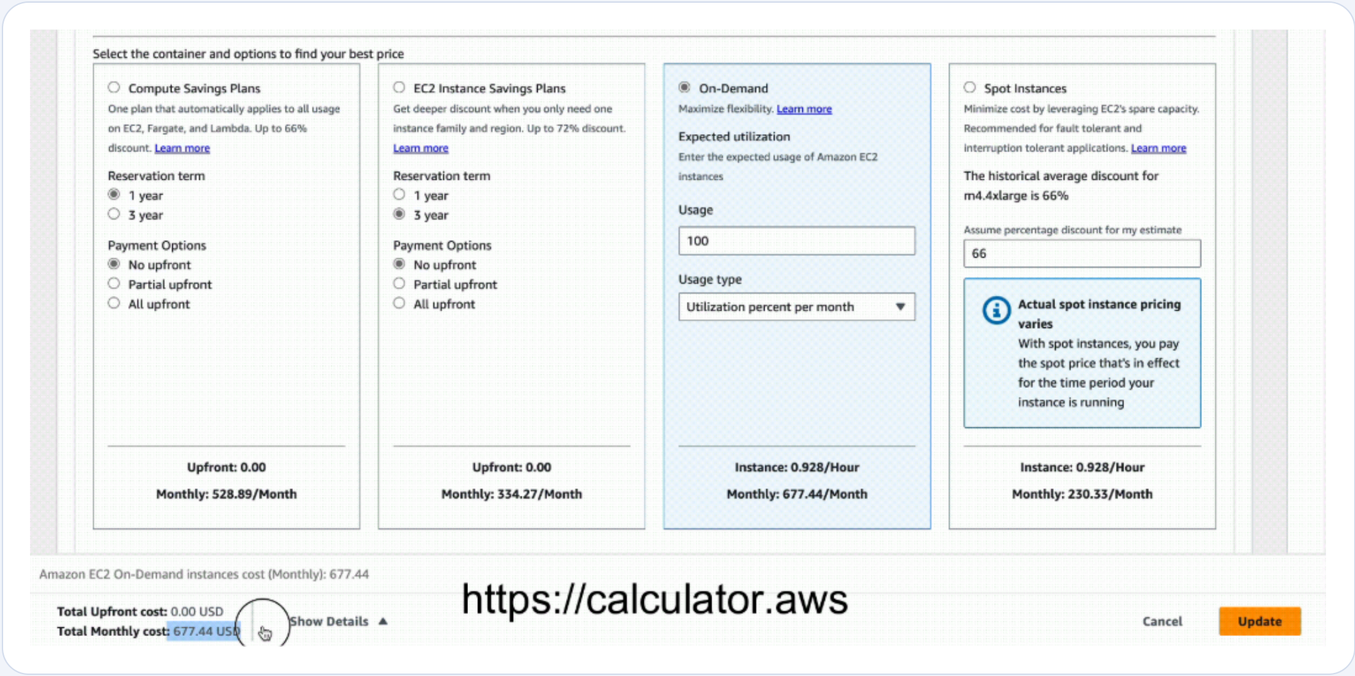
Estimate the cost of AWS products and services, and model your solutions before building them.

Explore the price points and calculations for your estimate, and find the instance types and contract terms that meet your needs.

You can plan your AWS costs and usage, or price out setting up a new set of instances and services.



AWS Pricing Calculator: comparing options



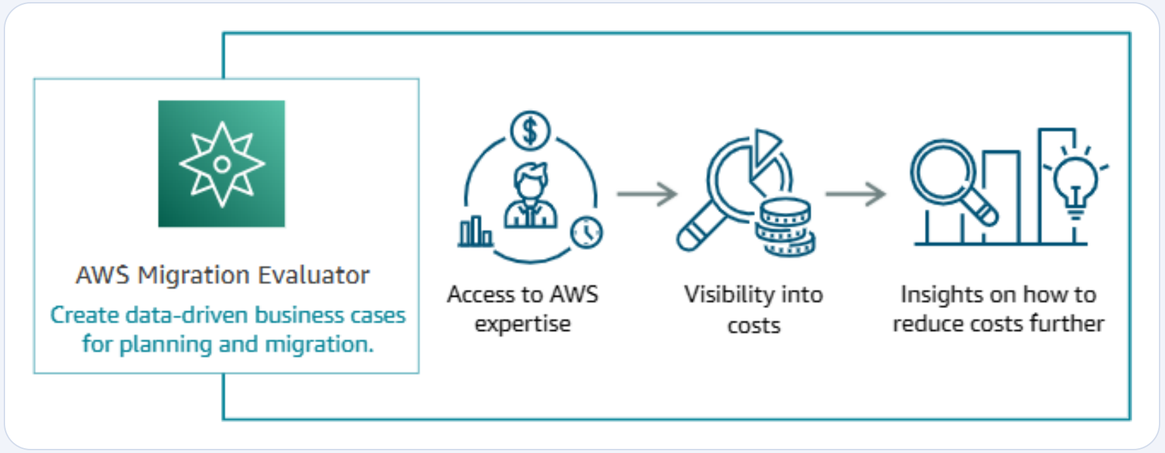
One workload, four prices: Savings Plans, On-Demand and Spot Instances side by side.

AWS Migration Evaluator

A complimentary service to create data-driven business cases for AWS Cloud planning and migration.

Creating business cases on your own can be time-consuming, and does not always identify the most cost-effective deployment and purchasing options.

You get access to AWS expertise, visibility into the costs of multiple migration strategies, and insights on reusing existing software licensing.



PART 3

Cost management tools

Cost Explorer, Budgets, the Billing Dashboard and Service Quotas.

Week 11 · 20

AWS cost management tools

AWS Billing console

Resources to manage your ongoing payments and payment methods registered to your AWS account



AWS Cost Management console

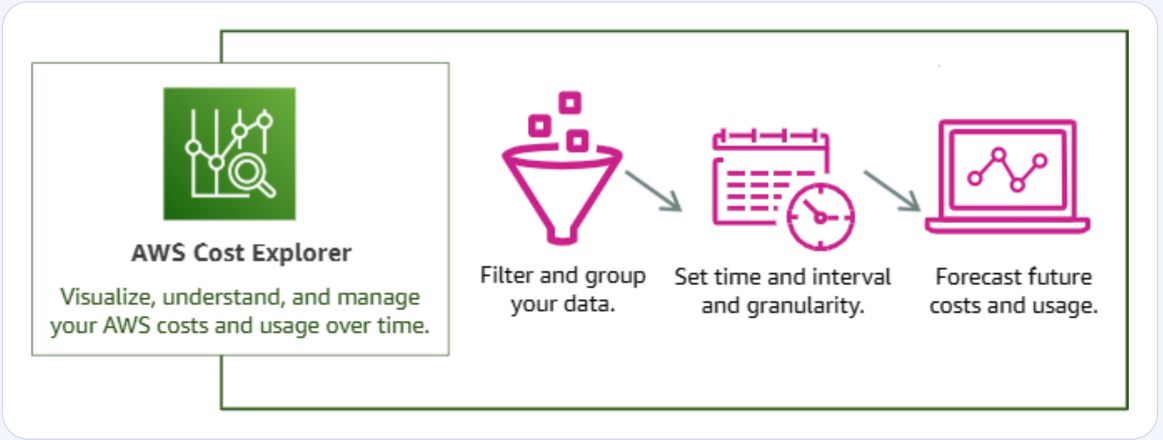
Features for budgeting and forecasting costs and methods for you to optimize your pricing to reduce your AWS bill

The Billing console manages payments. The Cost Management console budgets, forecasts and helps reduce the overall bill.

AWS Cost Explorer

View your AWS cost data as a graph, to visualize, understand and manage your costs and usage over time.

Analyze your data with filtering and grouping functions, at a monthly or daily level of granularity, and forecast your costs and usage.



AWS Cost Explorer benefits

See

View charts to see how your costs are being accumulated.

Look back

Review costs from the past 13 months to evaluate trends.

Look ahead

Forecast your costs over the next 3 months, based on current configurations and estimated usage.

Find problems

Discover how much you spend on AWS resources over time, and identify cost problem areas.

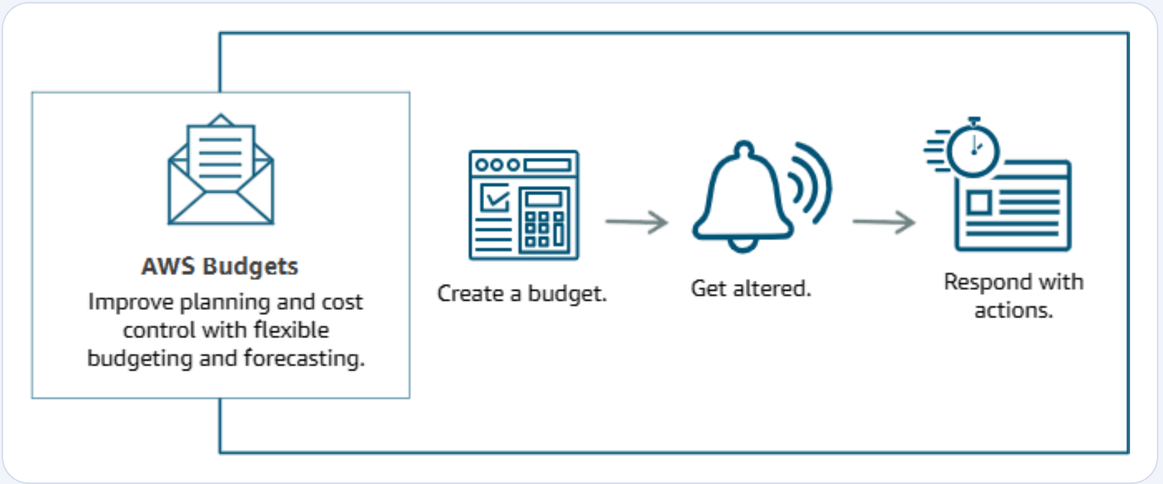
It also gives metrics such as which Availability Zones have the most traffic, or which linked AWS account is used the most.

AWS Budgets

Improves planning and cost control with flexible budgeting and forecasting, customized to how you want to budget your spending.

Receive daily, weekly or monthly reports. Automated alerts by email or Amazon SNS tell you when you exceed, or are forecasted to exceed, your budget.

You can also set up actions that run automatically, or through an approval process, when a budget target is exceeded.



AWS Billing Dashboard



View the status of your month-to-date AWS expenditure.



Identify the services that account for the majority of your overall expenditure.

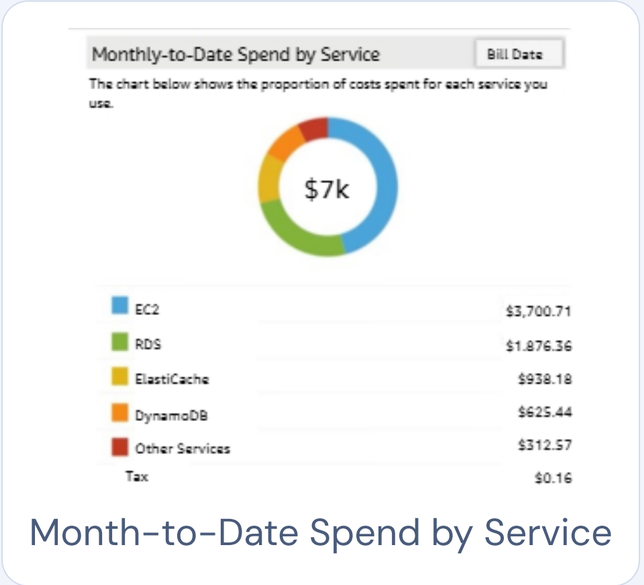


Understand at a high level how costs are trending.

AWS Billing Dashboard: two graphs



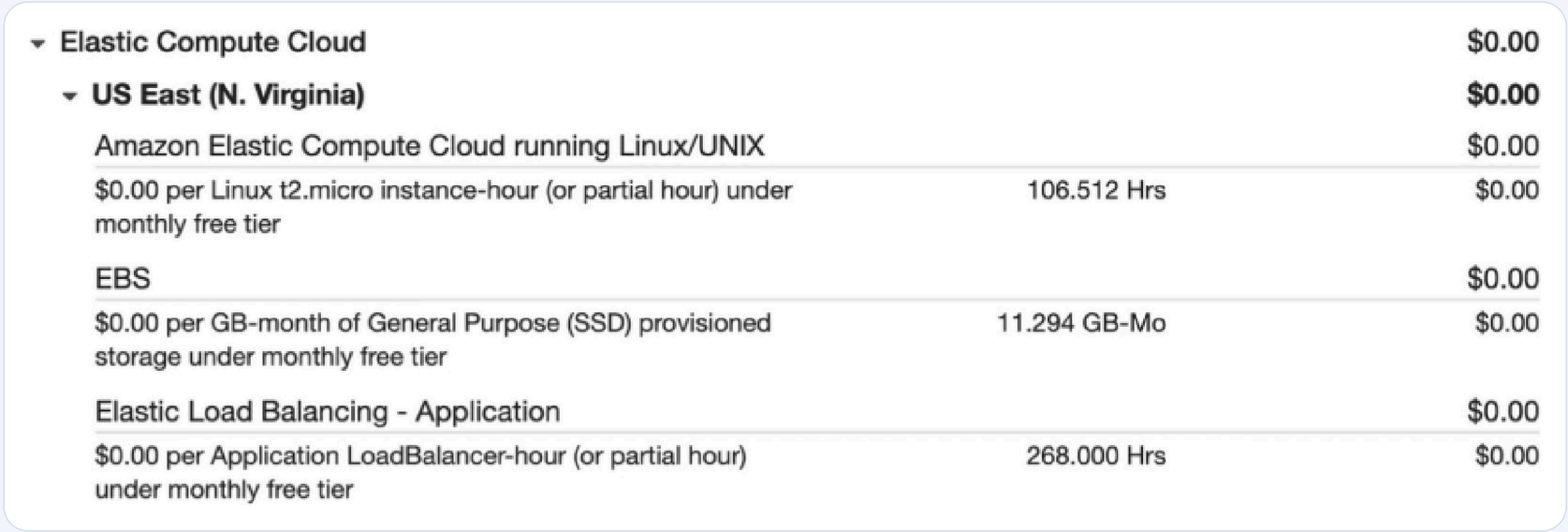
Spend Summary



Month-to-Date Spend by Service

Last month, month to date and the forecast on one side. The services you use most, and their share of the cost, on the other.

AWS Billing Dashboard example



Elastic Compute Cloud		**$0.00**
US East (N. Virginia)		**$0.00**
Amazon Elastic Compute Cloud running Linux/UNIX		$0.00
$0.00 per Linux t2.micro instance-hour (or partial hour) under monthly free tier	106.512 Hrs	$0.00
EBS		$0.00
$0.00 per GB-month of General Purpose (SSD) provisioned storage under monthly free tier	11.294 GB-Mo	$0.00
Elastic Load Balancing - Application		$0.00
$0.00 per Application LoadBalancer-hour (or partial hour) under monthly free tier	268.000 Hrs	$0.00

Each EC2 instance type, the EBS storage provisioned and the load balancer hours. All are under the Free Tier thresholds, so nothing is owed.

Paying less for Amazon EC2

Pay for running time

With Amazon EC2, you pay for only the compute time that you use while your instances are running.

Spot Instances

For a batch processing job that can withstand interruptions, Spot Instances can give up to 90 percent cost savings.

Commit

Find additional savings for Amazon EC2 by considering Savings Plans and Reserved Instances.

COST SCENARIO

What is your solution?

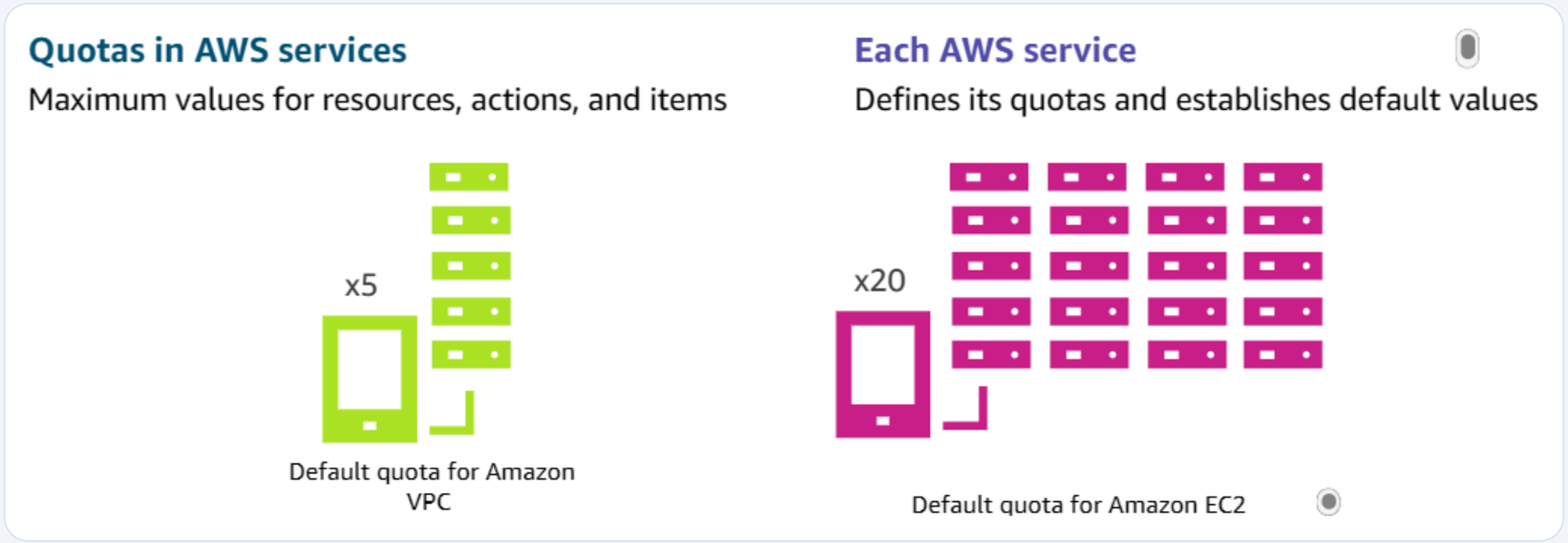
Your manager asks you to find where a new migration plan could cost less. Which AWS cost tool shows patterns in your spending over time and identifies cost problem areas?

Cost scenario: the answer

AWS Cost Explorer.

With AWS Cost Explorer, you can take advantage of filtering and grouping functions, get a better idea of what your costs and usage might look like in the future, and identify potential ways to reduce costs overall.

AWS Service Quotas



You can request a quota increase through the Service Quotas console, which manages quotas for many AWS services from one location.

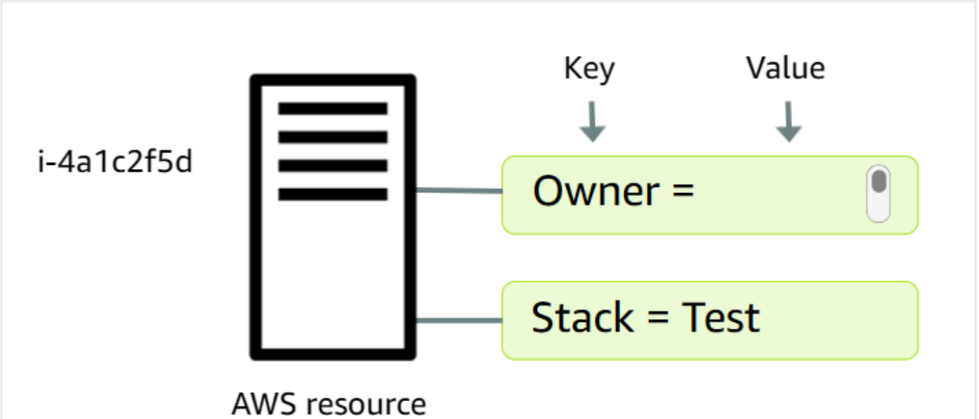
PART 4

Tagging and optimization

Know what each resource is for, then pay only for what the work needs.

Week 11 · 32

Tagging



Tags can help:

- Manage
- Identify
- Organize
- Search
- Filter

A tag key (for example, `CostCenter`, `Environment`, or `Project`). Tag keys are case sensitive.

A tag value (for example, `111122223333`, `Production`). Tag values are case sensitive.

A tag is a label of metadata on an AWS resource, made of a user-defined key and value.

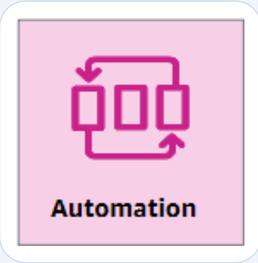
Common tagging strategies



Search and filter resources by tag, and group them with AWS Resource Groups and Tag Editor.



Break down costs by tag, such as cost center, customer or project, in Cost Explorer and billing reports.



Opt resources in or out of automated tasks, such as scripts that stop development environments after hours.

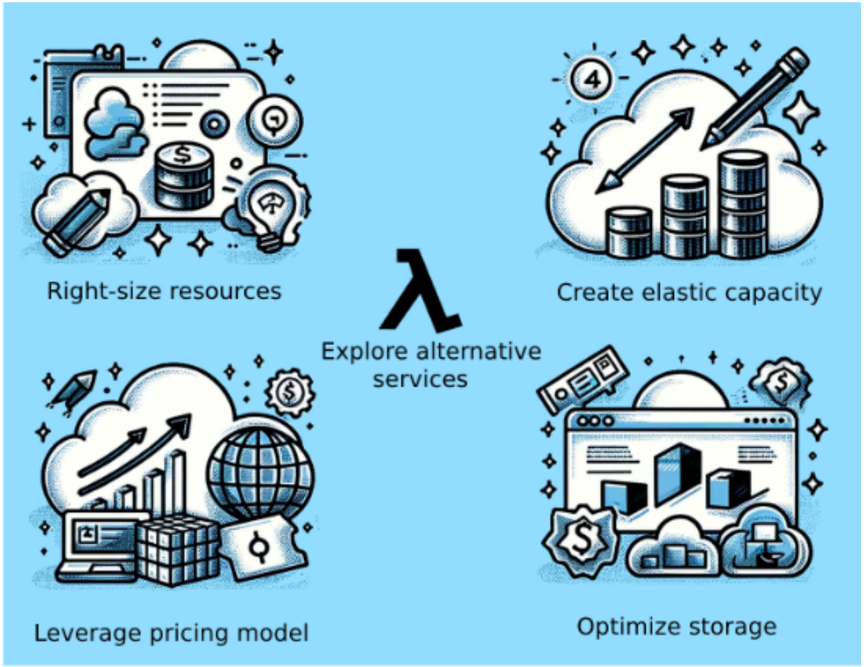


IAM policies support tag-based conditions, for example limiting EC2 API calls to one environment.

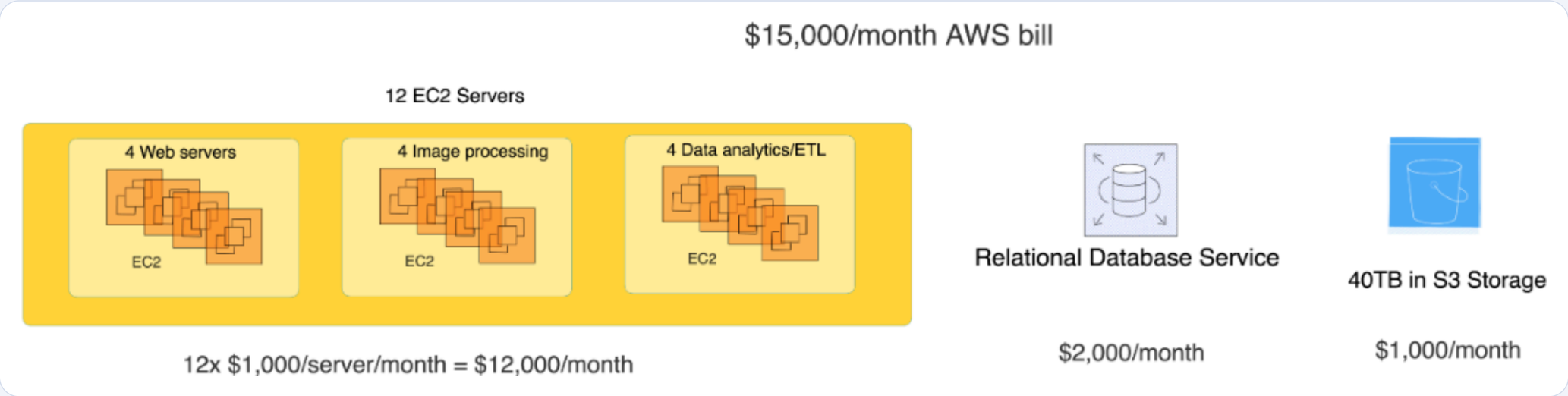
Cost optimization strategies

There are five steps to AWS cost optimization.

1. Right-size resources: for a 2-CPU task, do not use an 8-CPU server.
2. Create elastic capacity: run resources only when there is a business need.
3. Leverage pricing models to find additional savings.
4. Explore alternative services: Lambda for short-duration tasks.
5. Optimize storage: clean out unused storage.

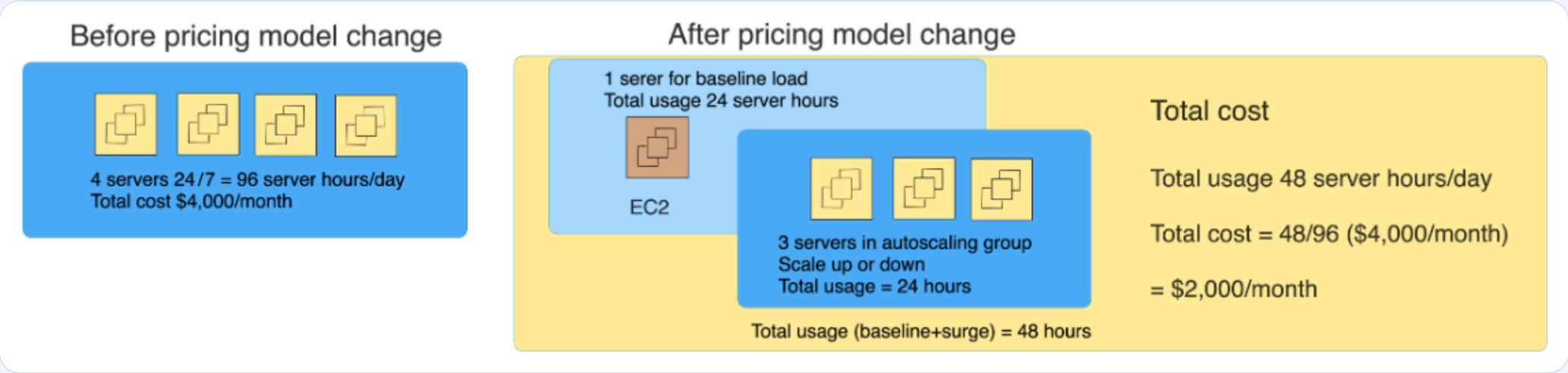


Optimization example



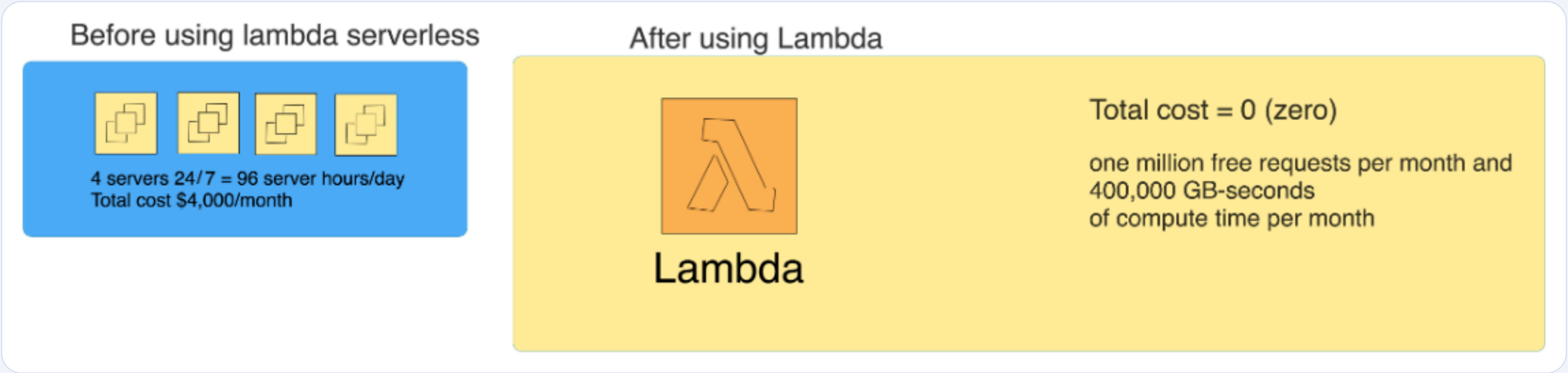
12 EC2 servers, a database service and S3 storage for $15,000 a month. Each task has four servers, sized for its peak usage.

Optimize size and capacity



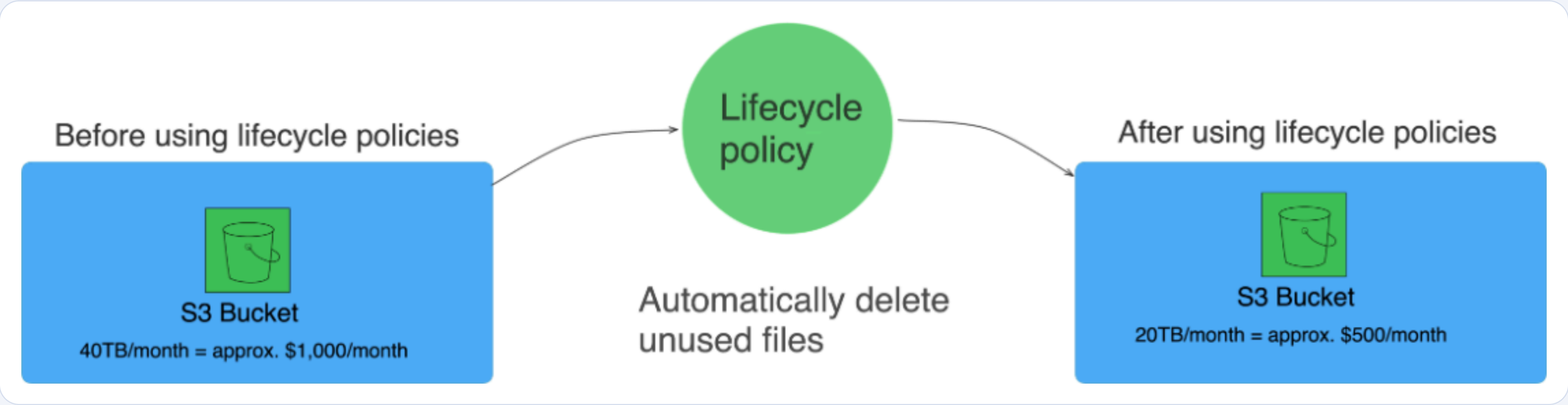
One server runs 24/7. Three surge servers add only 24 server-hours a day, so an Auto Scaling group saves 48 server-hours, or 50 percent.

Explore alternatives



Lambda suits short tasks such as image processing. Replacing 4 EC2 servers with Lambda saves $4,000 a month in server costs.

Optimize storage



Lifecycle policies are rules that define how files are purged. Auto-purge saves $500 a month, 50 percent of the storage costs.

Saving summary

Use case	Cost per month before	Cost per month after
Web servers	$4,000	$2,000
Analytics servers	$4,000	$222
Image processing servers	$4,000	$0
S3 storage costs	$1,000	$500
RDS database	$2,000	$2,000
Total	$15,000	$4,722

A 68.52 percent reduction. The savings you achieve will depend on your workload characteristics and business requirements.

Key takeaways

Pricing models

Pay for the individual services you need, for as long as you use them, with no long-term contracts.

Free Tier

Offers that are always free, offers that expire after 12 months, and short-term free trials.

TCO

Used to estimate and compare the direct and indirect costs of a product or a service.

Management Console

Everything you need to access and manage the AWS Cloud in one web interface.

Quotas

Resources, actions and items have quotas, many of which can be increased on request.

Tags

Tags help manage, identify, organize, search and filter resources.

Next session

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01

DevOps and IaC